Waste Classification — Data Detective Report

Hack[AI]Thon 2.0 · Round 1 · Online Screening | Team: [Your Team Name] | July 2026

1. Dataset Overview

Attribute	Value
Training images	445
Test images	115
Classes	cardboard, glass, metal, paper, plastic, trash
Baseline model	ResNet-18 (fixed, not modified)
Baseline accuracy	**42.53%** (random = 16.7%)

Class Distribution

Class	Count	Proportion
cardboard	82	18.4%
glass	78	17.5%
metal	84	18.9%
paper	86	19.3%
plastic	75	16.9%
trash	40	9.0%

Imbalance ratio: **2.15×** (trash most underrepresented)

2. Problems Found

2.1 Wrong Labels — 71.5% (318/445)

The majority of images are stored in incorrect class folders. Folder labels and filename-prefix labels disagree on most samples. The model learns **folder bias** (predicts "cardboard" for everything) rather than learning visual features of waste materials.

2.2 Duplicate Images

- **6 exact duplicate groups** — same image in multiple folders with different labels
- **11 near-duplicate pairs** — visually similar images with minor variations
- Causes train/validation leakage and inflated class counts

2.3 Blurry / Low-Quality Images

- **157 images (35.3%)** have Laplacian variance < 100
- Minimum variance: **0.6** (effectively blank)
- These images contribute noise instead of signal

2.4 Class Imbalance

- **trash** class has only **40 samples** vs 82–86 for others
- Model: **0% precision, 0% recall, 0% F1** on trash — class effectively invisible to the model

2.5 Outliers

- **22 file-size outliers** detected (range: 4.7 KB – 43 KB)
- Potential corrupted or anomalous samples

3. Evidence

Metric	Value
Overall validation accuracy	42.53%
Precision range	0.00 (trash) – 0.80 (cardboard)
Recall range	0.00 (trash) – 0.72 (cardboard)
Mislabeled images	318 (71.5%)
Blurry images	157 (35.3%)
Duplicate groups	6 exact + 11 near-duplicate

Key insight: Confusion matrix shows the model overwhelmingly predicts "cardboard" for all classes — evidence of **folder bias**. UMAP embeddings confirm no clean cluster separation by true class.

Evidence images available: 12 mislabel examples, 12 blurry examples, confusion matrix, UMAP plots, confidence histogram.

4. Suggested Fixes

Fix	Description	Expected Impact	Effort
1. Correct labels	Use filename-prefix labels + human review for ambiguous cases	+10–15% accuracy	High
2. Remove duplicates	Deduplicate exact + near-duplicate pairs	+3–5% accuracy	Low
3. Handle blurry images	Down-weight or remove low-quality samples	+2–3% accuracy	Low
4. Balance classes	Oversample/augment trash class	+5–8% trash recall	Medium
5. Stratified splits	Stratified sampling for train/val splits	Prevents leakage	Low
6. Self-training	Iterative pseudo-labeling with confidence threshold	+5–10% additional	Medium

5. Expected Impact

Primary Impact

- **Label correction** alone: +10–15 percentage points by removing dominant noise source
- **Class balancing**: trash recall from **0%** → **~30–40%**

Secondary Impact

- **Deduplication**: honest evaluation without data leakage
- **Blurry handling**: clean bottom 35% of dataset

Projected Combined Improvement

Baseline: 42.53% → Target: 60–70%

Recommendation

71.5% label noise is too severe for fully automated correction (Confident Learning only fixed 24/445 labels). **Human-in-the-loop verification** of ambiguous samples is essential. Remaining corrections via **self-training loop**: predict → filter confident ($p > 0.6$) → relabel → retrain.

Evidence: confusion matrix, UMAP embeddings, example images available in reports/figures/ and reports/evidence/